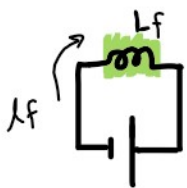
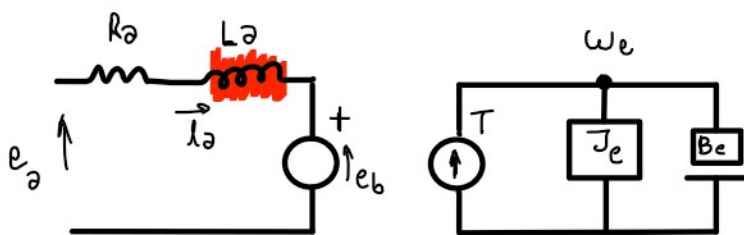
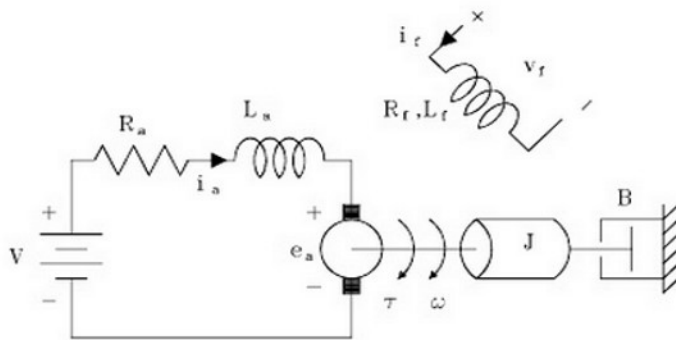


Motor DC: "Control por Armadura"



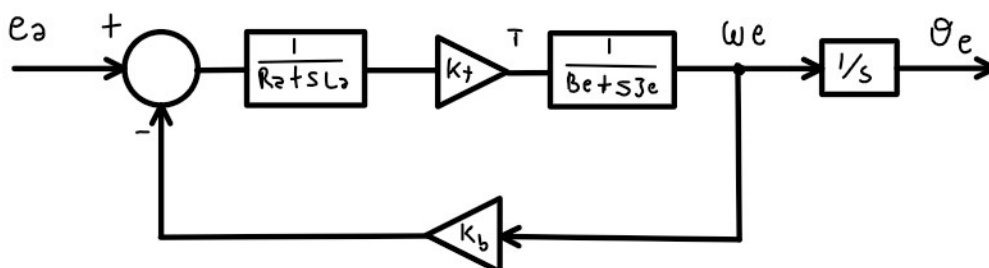
- Flujo Estático constante
- Se controla con $\langle i_a \rangle$ y $\langle e_a \rangle$

$$T = i_a \cdot K_t \quad K_t = K_b \text{ (Numericamente)}$$

$$e_b = \omega_e \cdot K_b \quad \dot{\theta}_e = \omega_e$$

$$-e_a + i_a R_a + L_a \dot{i}_a + e_b = 0$$

$$T = \dot{\omega}_e J_e + \omega_e \cdot B_e$$



$$u = e_a \quad y = \omega_e \quad x_1 = i_a \quad x_2 = \omega_e$$

$$u = L_a \dot{x}_1 + R_a x_1 + K \cdot x_2$$

$$\therefore \dots$$

$$\dot{X}_1 = \frac{1}{L_2} - \frac{R_2}{L_2} X_1 - \frac{K}{L_2} X_2$$

$$K X_1 = X_2 \cdot B_e + \dot{X}_2 J_e$$

$$\dot{X}_2 = \frac{K}{J_e} X_1 - \frac{B_e}{J_e} X_2$$

$$\begin{bmatrix} \dot{X}_1 \\ \dot{X}_2 \end{bmatrix} = \begin{bmatrix} -R_2/L_2 & -K/L_2 \\ K/J_e & -B_e/J_e \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} + \begin{bmatrix} 1/L_2 \\ 0 \end{bmatrix} [U]$$

$$[Y] = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} [U]$$

$$sI - A = \begin{pmatrix} s + R_2/L_2 & K/L_2 \\ -K/J_e & s + B_e/J_e \end{pmatrix}$$

$$1/z_e = R_2/L_2$$

$$1/z_m = B_e/J_e$$

$$(sI - A)^T = \begin{pmatrix} s + 1/z_e & -K/J_e \\ K/L_2 & s + 1/z_m \end{pmatrix}$$

$$z_m \gg z_e$$

La mecánica
es más lenta
que la electricidad

$$\begin{pmatrix} + & - \\ - & + \end{pmatrix}$$

$$\text{Adj}[(sI - A)^T] = \begin{pmatrix} s + 1/z_m & -K/L_2 \\ K/J_e & s + 1/z_e \end{pmatrix}$$

$$|sI - A| = (s + 1/z_e)(s + 1/z_m) + K^2/L_2 J_e$$

$$(sI - A)^{-1} = \begin{pmatrix} s + 1/z_m & -K/L_2 \\ K/J_e & s + 1/z_e \end{pmatrix} \frac{1}{(s + 1/z_e)(s + 1/z_m) + K^2/L_2 J_e}$$

$$H(s) = C \cdot (sI - A)^{-1} \cdot B + D$$

$$= (0 \ 1) \cdot \begin{pmatrix} \delta_{11} & \delta_{12} \\ \delta_{21} & \delta_{22} \end{pmatrix} \cdot \begin{pmatrix} 1/L_2 \\ 0 \end{pmatrix}$$

$$H(s) = \frac{W_e(s)}{E_a(s)}$$

$$H(s) = \frac{K}{(s + 1/2e)(s + 1/2a) L_2 J_e + K^2}$$

K determina los polos

↓